

Math 111: Calculus Homework 2

Due Friday, October 9 by 10:00P.M. via Gradescope

Covers Active Calculus §2.1–§2.3, §2.5, and §2.6. Problem 1 can be done now. Problems 2 and 4 use the chain rule (Monday, October 5), and Problem 3(c) uses derivatives of inverse functions (Wednesday, October 7).

Before you start. Write in sentences and paragraphs, as an explanation another student in this course could read and follow. Draw diagrams large and label them. You are encouraged to work with classmates; cite everyone you worked with, and write your solutions independently — duplicated solutions are an Honor Principle violation. Each problem is marked holistically on the 0–5 scale, with up to two further points for the quality of the writing. If you need an extension, use the [extension form](#) *before* the deadline.

1. A coffee cart, and where the money goes

A coffee cart has been open for a few months. Let $N(t)$ be the number of cups it sells per week, $P(t)$ the price of a cup in dollars, and $C(t)$ its total costs per week in dollars, all t weeks after opening. Ten weeks in, the owner knows:

$N(10) = 800$	$P(10) = 4.50$	$C(10) = 2400$
$N'(10) = -20$	$P'(10) = 0.10$	$C'(10) = 40$

- (a) Let $R(t) = N(t)P(t)$ be the weekly revenue. Find $R(10)$ and $R'(10)$, with units, and write one sentence saying what $R'(10)$ tells the owner.
- (b) The owner is surprised: the price is going *up*, so why isn't revenue? Draw the growing-rectangle picture from class for $R = N \cdot P$ at $t = 10$, and use it to explain the sign of $R'(10)$. Label which part of the picture is the effect of the price change and which is the effect of the change in sales, and give the dollar amount of each.
- (c) How large would $P'(10)$ have to be for revenue to be holding steady at week 10, if nothing else in the table changed? Then explain why “if nothing else changed” is doing a lot of work in that question: which other number in the table would you expect to move if the owner raised prices faster, and in which direction?
- (d) Let $A(t) = C(t)/N(t)$ be the cost of making one cup. Find $A(10)$ and $A'(10)$, with units. Compare $A'(10)$ with $P'(10)$, and say what the comparison means for the money the cart makes on each cup it sells.

2. Climbing into the cold

A hiker climbs a mountain. Let $a(t)$ be her altitude in feet, t hours after she starts, and let $T(a)$ be the air temperature in degrees Fahrenheit at altitude a feet. Suppose that on this mountain, today, $T'(a) = -0.0035$ for every altitude a on her route. The temperature *she experiences* at time t is $H(t) = T(a(t))$.

- (a) Say in a sentence what $T'(a) = -0.0035$ means, with units. Then suppose that at some moment she is climbing at 1200 feet per hour. How fast is the temperature she experiences changing at that moment? Explain why multiplying is the right thing to do, using the units and a sentence about what happens over the next few minutes of her climb — not by quoting the chain rule.

- (b) Her altitude is recorded every hour:

t (hours)	0	1	2	3	4
$a(t)$ (feet)	3000	4100	5300	6200	6700

Estimate $H'(2)$, with units. There is no formula for a or for H .

- (c) From the table alone, over which part of the hike does H' appear to be increasing, and over which part decreasing? Say what each means for the hiker, in a sentence about how cold it is getting.
- (d) A friend computes $H'(2)$ as $T'(2) \cdot a'(2)$. Explain what is wrong with writing $T'(2)$, and why, on *this* mountain today, the friend's number happens to come out right anyway. Then describe a situation — in words, about the weather — in which the friend's method would give the wrong answer.

3. Where the rules come from

- (a) We proved the power rule in class for positive whole-number exponents by expanding $(x + h)^n$. Use the quotient rule to show that it also holds for negative whole-number exponents: if n is a positive whole number, then $\frac{d}{dx}[x^{-n}] = -n x^{-n-1}$.
- (b) Now the exponent $\frac{1}{2}$. Assume that $s(x) = \sqrt{x}$ is differentiable for $x > 0$. Differentiate both sides of $(s(x))^2 = x$ with respect to x , and solve for $s'(x)$. Check that your answer is what the power rule predicts for $x^{1/2}$.
- (c) Suppose f is differentiable and has an inverse function f^{-1} , and that $f(3) = 5$ and $f'(3) = 4$. Find $(f^{-1})'(5)$. Then draw a picture that explains your answer: the graph of f near $(3, 5)$ with its tangent line, the graph of f^{-1} near $(5, 3)$ with its tangent line, and the line $y = x$. Say in a sentence why reflecting a line across $y = x$ turns its slope into the reciprocal.
- (d) The rules have hypotheses. Answer each of the following; give an example if one exists, and explain why not if it doesn't.
- Two functions f and g , *neither* of which is differentiable at $x = 0$, whose product fg is differentiable at $x = 0$.
 - Two functions f and g , both differentiable at $x = 0$, with $f'(0) = g'(0) = 0$ but $(fg)'(0) \neq 0$.

4. A rule that isn't, and a rule that is

A student submits the following.

Claim. For any differentiable function f , $\frac{d}{dx}[(f(x))^2] = 2f(x)$.

Argument. Write $u = f(x)$. The power rule says that the derivative of u^2 is $2u$. Substituting back, the derivative of $(f(x))^2$ is $2f(x)$.

- (a) Identify the *first* sentence of the argument that is wrong, and say precisely what is wrong with it.
- (b) Give a specific function f for which the claim is false, and show that it is false by computing both sides.

(c) State the correct rule for $\frac{d}{dx} \left[(f(x))^2 \right]$, explain where it comes from, and check it on your example from part (b).

(d) Your corrected rule gives a second route to the product rule. Check that

$$f(x)g(x) = \frac{1}{4} \left[(f(x) + g(x))^2 - (f(x) - g(x))^2 \right],$$

then differentiate the right-hand side using only your rule from part (c) and linearity. Show that the result is $f(x)g'(x) + g(x)f'(x)$.